Package 'nifdebug'

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Type Package

Title Debug Nonmem Input Format Data Sets Created With The NIF Package

Version 0.1.3

Description More about what it does (maybe more than one line).
Continuation lines should be indented.

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Imports dplyr, tidyr, ggplot2, nif, shiny, htmltools

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Suggests testthat (>= 3.0.0), tibble

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Depends R (>= 4.1.0)

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`debug_time_plot` *Interactive time metric comparison plot*

Description

This function creates an interactive Shiny plot to compare different time metrics (TIME, TAFD, TAD, NTIME) against each other. This is useful for debugging and verifying the consistency of time-related fields in a NIF data set. The x- and y-axis time metrics can be selected interactively via dropdown menus.

Usage

```
debug_time_plot(
  nif,
  analyte = NULL,
  dose = NULL,
  color = NULL,
  facet = "DOSE",
  min_time = NULL,
  max_time = NULL,
  size = 2,
  alpha = 0.5,
  log = FALSE,
  title = "Time metric relation",
  legend = TRUE,
  scales = "fixed",
  ...
)
```

Arguments

nif	A nif object.
analyte	The analyte(s) to be plotted, as character.
dose	The dose(s) to be filtered for.
color	The column(s) to be used for coloring.
facet	The column(s) to be used for faceting.
min_time	The minimal time in units of the selected time field, as numeric.
max_time	The minimal time in units of the selected time field, as numeric.
size	The size parameter to ggplot2::geom_point() as numeric.
alpha	The alpha parameter to ggplot2::geom_point(), as numeric.
log	Logarithmic y axis, as logical.
title	The plot title, as character. If none is provided, a generic title based on the analyte and the faceting will be chosen. Override this by setting title = "", if needed.
legend	Show the plot legend, as logical.
scales	The scales parameter to ggplot2::facet_wrap(), can be "fixed" (default), "free", "free_x" or "free_y".
...	Further parameters.

Value

A Shiny app object.

Examples

```
if (interactive()) {
  debug_time_plot(examplinib_poc_nif)
  debug_time_plot(examplinib_poc_nif, color = "ANALYTE")
  debug_time_plot(examplinib_poc_nif, analyte = "RS2023", facet = NULL)
}
```

`find_ex_source` *Map EX domain rows onto NIF administration records*

Description

For each row with 'EVID == 1', finds the corresponding SDTM **EX** record using Option A: subject, calendar interval ('EXSTDTC'–'EXENDTC' with imputed end when needed), 'EXTRT' (optionally mapped from 'ANALYTE'), and 'EXDOSE' matching 'DOSE'.

Usage

```
find_ex_source(ni, sdtm, analyte_extrt = NULL)
```

Arguments

`ni`	A 'nif' object.
`sdtm`	An 'sdtm' object with an 'ex' domain. If a 'dm' domain is present, 'RFENDTC' / 'RFSTDTC' are used when 'EXENDTC' or administration 'DTC' must be imputed.
`analyte_extrt`	Optional named character vector: names are 'ANALYTE' values in 'ni', values are the corresponding 'EXTRT' in **EX** (e.g. 'c(RS2023 = "RS2023")'). If 'NULL', 'ANALYTE' is compared to 'EXTRT' case-insensitively after trimming whitespace.

Details

Administration time is taken from 'DTC' when present; otherwise 'TRTDY' and 'RFSTDTC' from **DM** are used to approximate the calendar day (midnight in the timezone of 'RFSTDTC').

If 'SRC_DOMAIN', 'SRC_SEQ', and/or 'SRC_TESTCD' already exist, they are overwritten on administration rows and a warning is issued.

Value

'ni' with 'SRC_DOMAIN' ('"EX"'), 'SRC_SEQ' ('EXSEQ'), and 'SRC_TESTCD' ('EXTESTCD' if present in **EX**, otherwise 'NA') set on administration rows. Non-administration rows are left unchanged for these columns (except new columns are added as 'NA' where missing).

`nif_debug` *Interactive debug plot for NIF objects*

Description

Launches a Shiny app that renders an interactive version of [nif::plot.nif()]. The app includes a subject ('USUBJID') filter above the plot; the default is all subjects. Next to the selector, available subject fields from the NIF ('ID', 'SEX', 'AGE', 'WEIGHT', 'DL') are shown when a single subject is chosen. Data points are clickable; clicking a point displays the corresponding source SDTM record in a table below the plot.

Usage

```
nif_debug(
  nif,
  sdtm = NULL,
  analyte = NULL,
  dose = NULL,
  time = "TAFD",
  color = NULL,
  facet = "DOSE",
  min_time = NULL,
  max_time = NULL,
  cfb = FALSE,
  dose_norm = FALSE,
  log = TRUE,
  lines = TRUE,
  size = 2,
  alpha = 1,
  scales = "fixed",
  base_size = 16
)
```

Arguments

nif	A nif object. If it contains ‘SRC_DOMAIN‘ and ‘SRC_SEQ‘, clicking points enables SDTM source lookup and will display the SDTM tables.
sdtm	An optional ‘sdtm‘ object used to look up source records. If ‘NULL‘ (or missing the required NIF source fields), the SDTM/EX tables are hidden.
analyte	The analyte(s) to be plotted, as character.
dose	The dose(s) to be filtered for.
time	The time field for the x-axis. One of ’TIME’, ’NTIME’, ’TAFD’ or ’TAD’.
color	The column(s) to be used for coloring.
facet	The column(s) to be used for faceting.
min_time	The minimal time, as numeric.
max_time	The maximal time, as numeric.
cfb	Plot change from baseline, as logical.
dose_norm	Dose-normalized values, as logical.
log	Logarithmic y axis, as logical. Defaults to ‘TRUE‘ (Log checked in the app).
lines	Plot lines, as logical.
size	The ‘size‘ parameter to [ggplot2::geom_point()], as numeric.
alpha	The ‘alpha‘ parameter to [ggplot2::geom_point()], as numeric.
scales	The ‘scales‘ parameter to [ggplot2::facet_wrap()].
base_size	Base font size as numeric.

Details

Source lookups are only available when: - ‘sdtm‘ is provided, and - the NIF includes ‘SRC_DOMAIN‘ and ‘SRC_SEQ‘.

If SDTM lookup is not available, the upper SDTM/EX tables are not shown, but the NIF record table can still be displayed.

‘SRC_DOMAIN‘ and ‘SRC_SEQ‘ are typically present when the NIF is built with ‘debug = TRUE‘ or when the global ‘nif::nif_option("debug")‘ is set.

Value

A Shiny app object.

See Also

[nif::plot.nif()], [nif::nif_viewer()]

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